

SKINSIGHTS

Presentation by
Bridget Walsh, Rithika Ravichandran, Kathryn Tran



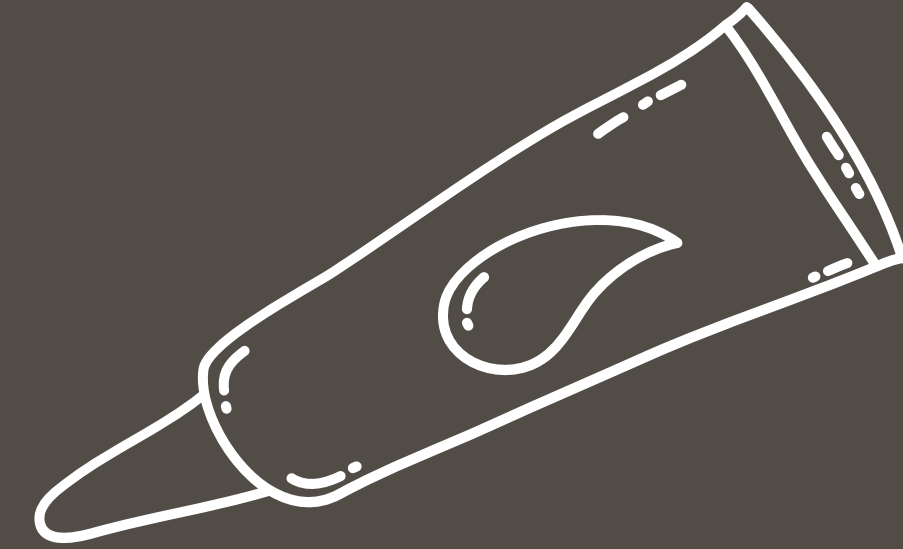
Introduction

What is SKINSIGHTS

SKINSIGHTS is a **skincare database application** that connects users to the products in their routines and the ingredients inside them. It helps people see which products are safe for their skin type, which ingredients may cause irritation, and how brands compare in terms of safety.



Objectives



Problem:

- Consumers **struggle to understand ingredient lists** and which products are safe for their skin type.
- A product that works for one person may irritate another due to **sensitivities, allergies, or conditions**.
- Users often don't know **which ingredients to avoid** or which **brands are safest for them**.

Goals:

Our database aims to help users:

- Identify **what ingredients** are in the products they use
- See which products other users with **similar skin types prefer**
- Understand **which ingredients are safe or unsafe** for their skin type
- Compare brands and stores based on **ingredient safety**





Data Sources and Methods



Data Collection

We distributed a Google Form to collect real user inputs, including skin type, common conditions (sensitivity, acne, eczema, rosacea), allergies, and each user's top three most frequently used skincare products, along with their brands.

We then compiled ingredient lists for the selected products using a verified cosmetic ingredient database. To ensure accuracy, we standardized ingredient naming across brands and variations.

Data Cleaning

- Created separate tables for brands and products for normalization
- Removed users who filled out the form incorrectly or with incomplete data
- Standardized ingredient names to ensure consistency
- Limited the dataset to skincare products only (no makeup or unrelated items)

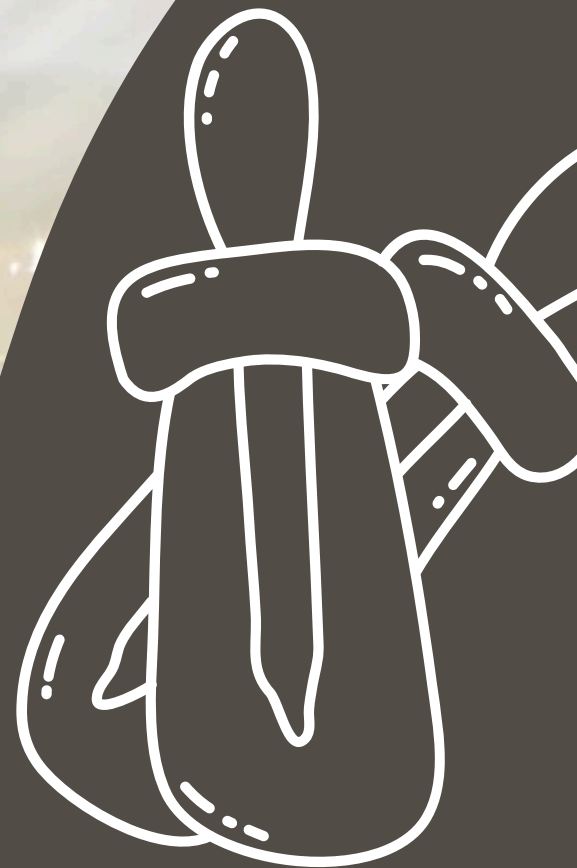
Tables Used

Core Tables

- Users
- Products
- Ingredients
 - Brands
- Locations
 - Safety

Join Tables

- Users_Products
- Products_Ingredients
 - Users_Brands
- Location_Products



01.

What is the most common brand used by users with dry skin?

02.

Which user has the most products, and what are the product types, brands, and expense levels?

03.

What ingredients are associated with oily skin (safe and unsafe)?

04.

Which products are safe for all skin types?

09.

What are the top five ingredients unsafe for the greatest number of skin types?

05.

For acne-prone users, how many of their products contain unsafe ingredients?

06.

Which brands contain the highest number of unsafe ingredients?

07.

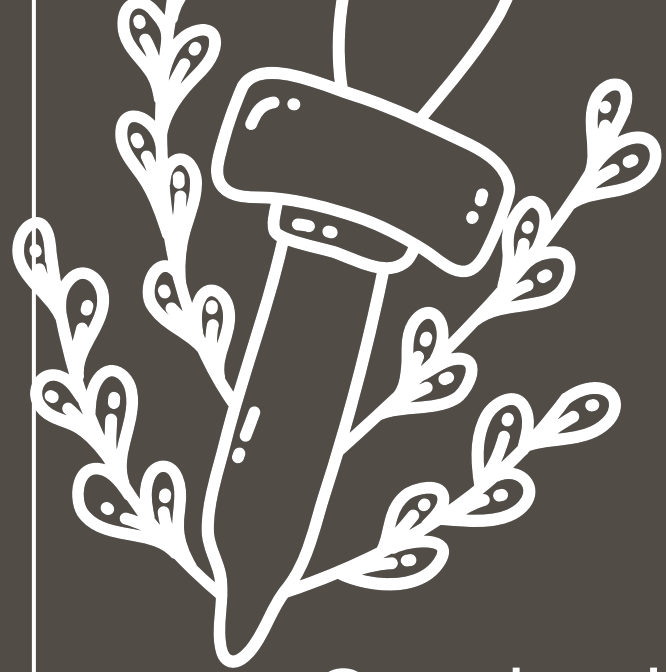
Which brands use the fewest unsafe ingredients (the "cleanest" brands)?

08.

Which store location sells the highest number of high-risk products?



Queries



Database Design

Our database includes users, products, ingredients, brands, and store locations.

Users connect to products through a many-to-many relationship. Products contain multiple ingredients, and ingredients may appear in several products. Brands offer many products, and locations may sell products from multiple brands.

A safety table links ingredients to skin-type safety guidelines. These relationships allow SKINSIGHTS to map routines to formulas, ingredient risks, and product availability.

Entities

- Users — skin type, sensitivities, allergies
- Products — product name, type, brand
- Ingredients — ingredient list + ID
- Brands — brand name + expense level
- Locations — stores where products are sold
- Safety — ingredient safety by skin type
- Join Tables — connect users to products, products to ingredients, and brands/locations

Relationships

- Users ↔ Products
- Products ↔ Ingredients
- Users ↔ Brands
- Products ↔ Locations
- Safety ↔ Ingredients

Insights and Results

Our queries revealed key insights about ingredient safety, brand behavior, and user routines. We identified which products are safe, which brands use the most high-risk ingredients, and how stores differ in what they carry. These findings show how SKINSIGHTS can help users make more informed skincare choices.

Top 5 Ingredients Unsafe for the Greatest Number of Skin Types



Ingredient & Product Safety

- Identified all safe vs. unsafe ingredients for oily skin
- Only two products were safe for every skin type.
- The most widely used unsafe ingredients included Fragrance, Alcohol Denat, and SLS/SLES.

Brand & Store Insights

- CeraVe and Cetaphil had the highest number of unsafe ingredients overall.
- Cleanest brands were Aquaphor, PanOxyl, and Numbuzin.
- Target carried the most high-risk products compared to Sephora and Walmart.

User Behavior Patterns

- For dry-skin users, the top brands used were CeraVe, Neutrogena, and The Ordinary.
- We identified which user used the most products and analyzed their routine.
- Two acne-prone users were unknowingly using products with unsafe ingredients.

Limitations & Future Work

Limitations

- Limited number of products and brands included in the database
 - Small user sample size from the Google Form
 - Safety classifications do not cover every skin concern or ingredient variation
 - Ingredient lists rely on manually collected data, which may contain inconsistencies

Future Work

- Expand the database to include more products, brands, and product categories
 - Collect more user data for stronger pattern detection
 - Automate ingredient extraction from product websites or APIs
 - Improve the safety scoring model by including ingredient concentrations
 - Create a front-end interface to help users explore routine safety in real time

Conclusion

- We built a functioning skincare database using real user responses that mimics an actual personalized skincare app.
- Users can find products that match their skin type, allergies, and conditions, and filter out unsafe or unwanted ingredients and brands.
- Our database structure accurately maps real relationships between products, ingredients, brands, users, and store locations.
- The queries demonstrate how the application can provide meaningful insights, from ingredient safety to brand comparisons.
- Limitations included manually importing large amounts of data and difficulties syncing CSV imports across group members.
- Overall, our database achieved what we set out to do and shows strong potential as a real-world skincare tool.



References

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Thank You